In [11]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
```

In [14]:
```python
df_train = pd.read_csv("data/processed/monthly_housing_train.csv")
df_train["date"] = pd.to_datetime(df_train["date"])
df_train = df_train.set_index("date")
df_train.head(3)
```

Out[14]:

	hpi	numsold
date		
1991-01-01	100.00	30
1991-02-01	100.48	40
1991-03-01	100.74	51

Visualize train data

In [15]:
```python
plt.figure(figsize=(10, 4))

plt.plot(
    df_train.index,
    df_train["hpi"],
    linewidth=0.8
)

plt.title("Housing Price Index Over Time (Training Set)")
plt.xlabel("Date")
plt.ylabel("HPI")
plt.grid(True, alpha=0.3)

ax = plt.gca()
ax.xaxis.set_major_locator(mdates.YearLocator(4))   # 每 4 年一格
ax.xaxis.set_major_formatter(mdates.DateFormatter("%Y"))

plt.tight_layout()
plt.show()
```

Summary Statistic

In [16]:
```python
summary_stats = pd.DataFrame({
    "HPI": [
        df_train["hpi"].mean(),
        df_train["hpi"].std(),
        df_train["hpi"].min(),
        df_train["hpi"].quantile(0.25),
        df_train["hpi"].median(),
        df_train["hpi"].quantile(0.75),
        df_train["hpi"].max(),
        df_train["hpi"].std() / df_train["hpi"].mean()
    ],
}, index=[
    "Mean",
    "Standard deviation",
    "Minimum",
    "First quartile",
    "Median",
    "Third quartile",
    "Maximum",
    "CV"
])

summary_stats_rounded = summary_stats.round(3)
summary_stats_rounded
```

Out[16]:

	HPI
Mean	157.199
Standard deviation	42.155
Minimum	100.000
First quartile	116.117
Median	154.290
Third quartile	195.122
Maximum	228.710
CV	0.268